



DEPARTAMENTO
DE INFORMÁTICA
PUC-RIO

Software &
AI Engineering Lab



Research Problem: A Statistical Approach to Assess the Functional Correctness of ML-Based AI Systems

Case Study (A)

December, 16 - 2025

Lean Research Inception

Workshop

1. Workshop

1.1. Problem Vision Alignment

1.2. Research Problem Formulation

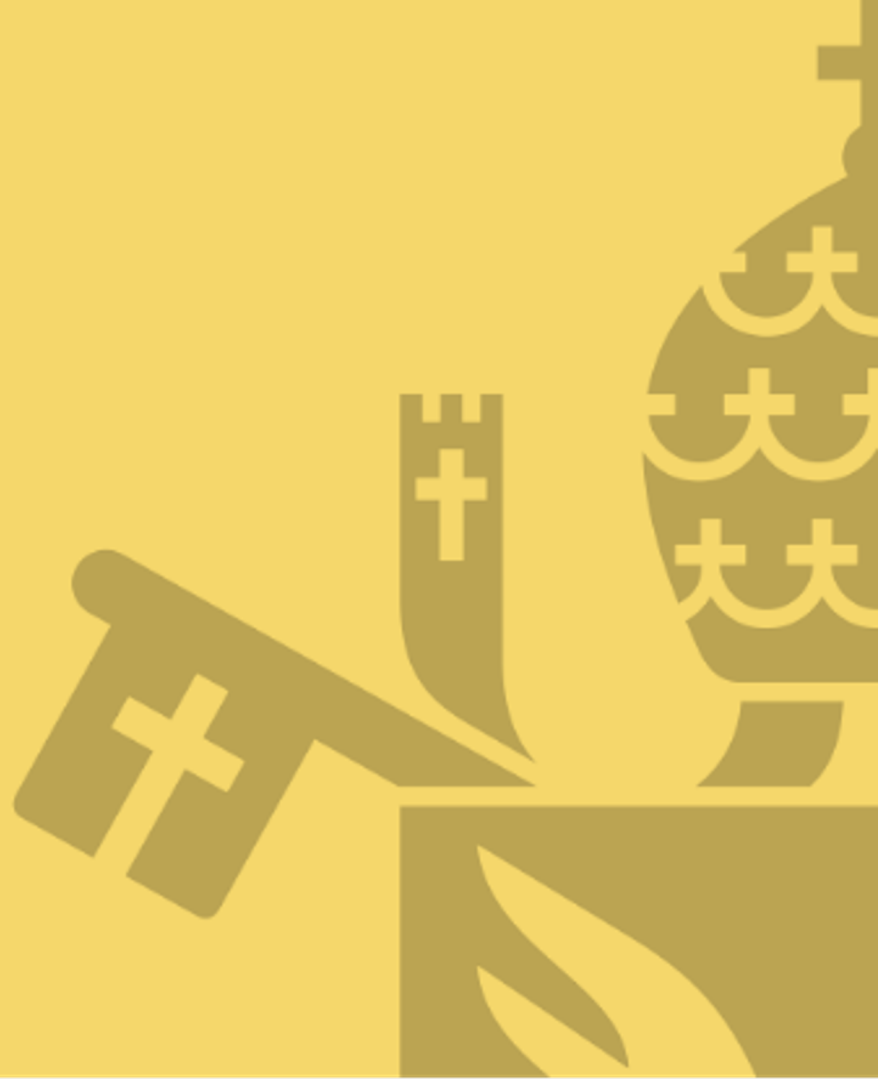
1.3. Research Problem Assessment

1.4. Go / Pivot / Abort Decision

Workshop

Approximately:

1 hour and 30 minutes to 2 hours.



Phase 2

Problem Vision Alignment

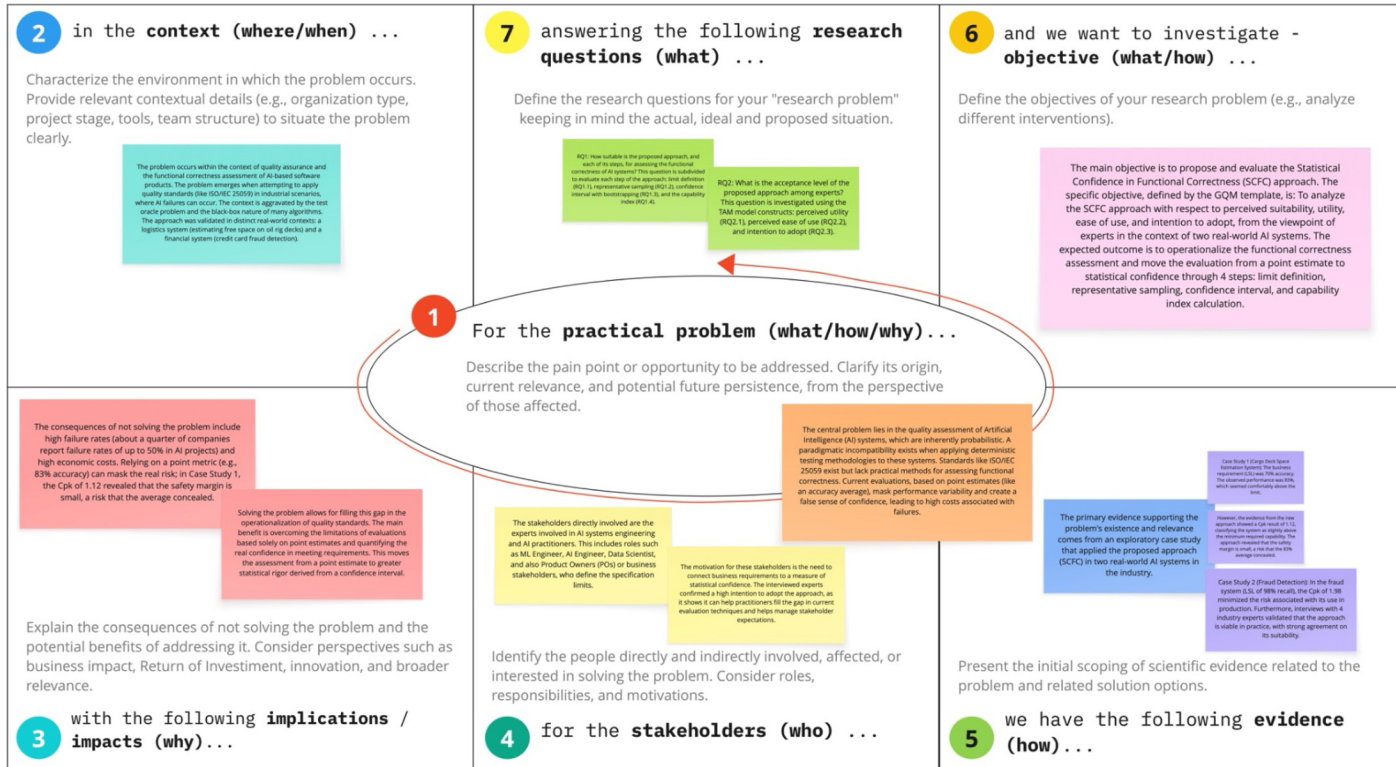


For this activity we have:

01:00:00

Starting from the **draft of the research problem** presented, we will work on **discussing** and **refining each** of its **seven attributes**.

Problem Vision - Albertini



Download
the Problem
Vision Draft
prepared by
the
Researcher

Miro Template Link: https://miro.com/app/board/uXjVJr0vCsY=

Phase 3

Research Problem Formulation



For this activity we have:

00:30:00

Let's **organize** our **ideas** and **rewrite the research problem** in an organized and structured way to facilitate reading and evaluation in the next step.

Upon rereading, we can discuss whether the problem is clear and reflects the views of all participants.

3 - Research Problem Formulation

For the practical problem involving the lack of approaches to evaluate and ensure the functional correctness of Artificial Intelligence (AI) systems, **within the context** of applying quality standards, such as **ISO/IEC 25059**, in industrial settings,

which bring the following implications/impacts: accumulation of technical debt and rework; for example, high failure rates (approximately one quarter of companies report failure rates of up to 50% in AI projects); high economic costs; reliance on a single point estimate (e.g., 83% accuracy) that may mask the actual risk. Addressing this problem can enable the operationalization of quality standards, with the main benefit being the ability to overcome the limitations of evaluations based solely on point estimates and to quantify real confidence in meeting requirements. This shifts the assessment from a point estimate to a higher level of statistical rigor derived from a confidence interval.

For the stakeholders: AI systems engineering specialists, Machine Learning Engineers, AI Engineers, Data Scientists, as well as Product Owners (POs) and Business Owners.

We have the following evidence: functional correctness is currently evaluated based on a point estimate. For example: (i) *Cargo Deck Free Space Estimation System*: the business requirement is 70% accuracy (stakeholder specification limit); (ii) *Credit Card Fraud Detection System*: recall above 98% (stakeholder specification limit).

3 - Research Problem Formulation

We aim to propose and evaluate a *Statistical Confidence in Functional Correctness (SCFC)* approach with respect to perceived adequacy, usefulness, ease of use, and intention to adopt, from the perspective of experts in real-world AI system contexts. The expected outcome is to operationalize the evaluation of functional correctness by moving from a point estimate to statistical confidence through four steps: (1) definition of specification limits, (2) representative data sampling, (3) confidence interval estimation, and (4) calculation of the capability index.

Answering the following research questions:

RQ1: How adequate is the proposed approach and each of its steps for evaluating the functional correctness of AI systems?

RQ1.1: Definition of metrics and specification limits; **RQ1.2:** Representative data sampling; **RQ1.3:** Confidence interval estimation using bootstrapping; and **RQ1.4:** Calculation of the *capability index* based on the outcomes of the previous RQs.

RQ2: What is the level of acceptance of the proposed approach among experts?

RQ2.1: Perceived usefulness; **RQ2.2:** Perceived ease of use; and **RQ2.3:** Intention to adopt.

Phase 4

Research Problem Assessment



For this activity we have:

00:15:00

We will use the **semantic differential scale** to **evaluate** the **formulated research problem**.
We will do this **individually**.

* Each participant will use one of the following slides for this activity.

Participant:

P1

Role:
Researcher

Individual - Relevance Assessment Scale

Valuable: To what extent does addressing the formulated problem have the potential to generate meaningful value for industrial practice (e.g., *business impact, measurable benefits, return of investment, or innovative contributions*)?

1. 2. 3. 4. 5. 6. 7.

Not Valuable

☐☐☐☐☐☐☒

Valuable

Comments / Suggestions:

This approach is valuable because it directly addresses a technical debt problem. It generates value by exposing risks that point metrics hide, providing greater statistical confidence through the confidence interval.

Applicable: To what extent can the expected results of this formulated research problem be realistically applied in real industry scenarios (considering factors such as *adoption potential, contextual fit, and stakeholder willingness to use the outcomes*)?

1. 2. 3. 4. 5. 6. 7.

Not Applicable

☐☐☐☐☐☐☒

Applicable

Comments / Suggestions:

I believe this approach directly connects business requirements with technical metrics, facilitating communication between project stakeholders.

Feasible: To what extent can this research problem be realistically investigated with the resources typically available (e.g., *time, data, access to practitioners, tools, and expertise*)?

1. 2. 3. 4. 5. 6. 7.

Not Feasible

☐☐☐☐☐☐☒

Feasible

Comments / Suggestions:

The approach is structured in 4 clear and low-complexity steps, demonstrating an accessible and replicable assessment.

Participant:

P2

Role:
Practitioners

Individual - Relevance Assessment Scale

Valuable: To what extent does addressing the formulated problem have the potential to generate meaningful value for industrial practice (e.g., *business impact, measurable benefits, return of investment, or innovative contributions*)?

1. 2. 3. 4. 5. 6. 7.

Not Valuable

☐☐☐☐☐☐☒

Valuable

Comments / Suggestions:

With regard to improving the quality and reliability of AI systems, this tool is very useful for ensuring product confidence and team morale.

Applicable: To what extent can the expected results of this formulated research problem be realistically applied in real industry scenarios (considering factors such as *adoption potential, contextual fit, and stakeholder willingness to use the outcomes*)?

1. 2. 3. 4. 5. 6. 7.

Not Applicable

☐☐☐☐☐☒☐

Applicable

Comments / Suggestions:

About Applicable: Depending on the client's and project's maturity, some requirements may simply not be known at certain stages of the project or may simply change over time. However, overall, it is clearly very important for the stakeholder by providing predictability to the project.

Feasible: To what extent can this research problem be realistically investigated with the resources typically available (e.g., *time, data, access to practitioners, tools, and expertise*)?

1. 2. 3. 4. 5. 6. 7.

Not Feasible

☐☐☐☐☐☐☒

Feasible

Comments / Suggestions:

No comments

Participant:

P3

Role:
Practitioners

Individual - Relevance Assessment Scale

Valuable: To what extent does addressing the formulated problem have the potential to generate meaningful value for industrial practice (e.g., *business impact, measurable benefits, return of investment, or innovative contributions*)?

1. 2. 3. 4. 5. 6. 7.

Not Valuable

☐☐☐☐☐☒☐

Valuable

Comments / Suggestions:

Yes, because the main consequence of a poorly defined requirement is rework. And rework can be directly defined as cost.

Applicable: To what extent can the expected results of this formulated research problem be realistically applied in real industry scenarios (*considering factors such as adoption potential, contextual fit, and stakeholder willingness to use the outcomes*)?

1. 2. 3. 4. 5. 6. 7.

Not Applicable

☐☐☐☐☒☐☐

Applicable

Comments / Suggestions:

Because the approach seems simple, it is highly applicable.

Feasible: To what extent can this research problem be realistically investigated with the resources typically available (e.g., *time, data, access to practitioners, tools, and expertise*)?

1. 2. 3. 4. 5. 6. 7.

Not Feasible

☐☐☐☐☒☐☐

Feasible

Comments / Suggestions:

I believe that obligation is directly linked to the learning curve.

Participant:

P4

Role:
Practitioners

Individual - Relevance Assessment Scale

Valuable: To what extent does addressing the formulated problem have the potential to generate meaningful value for industrial practice (e.g., *business impact, measurable benefits, return of investment, or innovative contributions*)?

1. 2. 3. 4. 5. 6. 7.

Not Valuable

☐☐☐☐☒☐☐

Valuable

Comments / Suggestions:

The proposed approach generates practical value by operationalizing these quality standards, offering a structured path to start from metrics and limits defined at a general level and, from there, specialize the evaluation for specific business contexts and requirements. However, the proposal could be strengthened by including guiding metric models for different types of machine learning problems (e.g., classification, regression, clustering, and time prediction). Such an addition would facilitate the adaptation of the approach by potential users, allowing for better alignment with their application contexts without compromising the generality of the method.

Applicable: To what extent can the expected results of this formulated research problem be realistically applied in real industry scenarios (considering factors such as *adoption potential, contextual fit, and stakeholder willingness to use the outcomes*)?

1. 2. 3. 4. 5. 6. 7.

Not Applicable

☐☐☐☐☐☐☒

Applicable

Comments / Suggestions:

The expected results are realistically applicable in industrial settings, as the proposed approach aligns with existing quality standards (e.g., ISO/IEC 25059) and integrates naturally with current workflows for developing and evaluating ML systems. By relying on statistical concepts already familiar to industry professionals (such as confidence intervals and capability analysis), the approach has high adoption potential. Furthermore, its flexibility allows for adaptation to different organizational contexts and use cases, increasing stakeholder willingness to apply the results in real-world scenarios without requiring disruptive changes to established processes.

Feasible: To what extent can this research problem be realistically investigated with the resources typically available (e.g., *time, data, access to practitioners, tools, and expertise*)?

1. 2. 3. 4. 5. 6. 7.

Not Feasible

☐☐☐☐☐☐☒

Feasible

Comments / Suggestions:

The research problem can be realistically investigated with typically available resources, since the proposed approach uses evaluation data already generated in the normal AI systems development cycle, without requiring specialized infrastructure or tools beyond those commonly adopted in the industry. The time required is compatible with medium-sized experimental projects, and access to experts is facilitated by the participation of professionals directly involved in ML engineering and quality assurance activities. Furthermore, the statistical methods employed are well-established and supported by widely used libraries, making the study feasible in both academic and industrial environments.

Phase 5

Go / Pivot / Abort Decision



For this activity we have:

00:30:00

We will **consolidate** the **group's assessment**, **discuss** everyone's perception of the formulated research problem, and **define** the **next step**.

Assessment Results

Relevant Research Problem (RRP) - Assessment

	1.	2.	3.	4.	5.	6.	7.	Median
Valuable: To what extent does addressing the formulated problem have the potential to generate meaningful value for industrial practice (<i>e.g., business impact, measurable benefits, return of investment, or innovative contributions</i>)?								
Not Valuable					X		Valuable	6,5
Applicable: To what extent can the expected results of this formulated research problem be realistically applied in real industry scenarios (<i>considering factors such as adoption potential, contextual fit, and stakeholder willingness to use the outcomes</i>)?								
Not Applicable					X		Applicable	6,5
Feasible: To what extent can this research problem be realistically investigated with the resources typically available (<i>e.g., time, data, access to practitioners, tools, and expertise</i>)?								
Not Feasible					X		Feasible	7

5 - Go / Pivot / Abort Decision

Comments / Observations

Decision: Go

The group assesses the problem as valuable, applicable, and feasible, with strong alignment to industrial needs and the use of well-established methods. The observations indicate incremental adjustments (e.g., metric guidelines and mitigation of the learning curve) to strengthen adoption, without the need to pivot or abort the proposal.



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